MIL-A-12831D (MU)
11 July 1973
SUPERSEDING
MIL-A-12831C (MU)
14 May 1965

MILITARY SPECIFICATION

ACTIVATOR, M2
PARTS FOR

1. SCOPE

1.1 This specification covers the parts for one type of activator designated as Activator, M2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1345

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DRAWINGS

U. S. ARMY MUNITIONS COMMAND

7548048 - Activator, AT, Mine, M2.
9207815 - Activator, M2, Parts Arrangement.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Plastic parts.-The plastic parts shall comply with all requirements specified on Drawing (dwg.) 9207815, and with all requirements in applicable specifications.

3.3 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, porosity, warpage, checks, chipped edges, flashing, chips, dirt, grease, rust and other foreign matter, and defects which would affect their serviceability and appearance. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.4 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to

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MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material pruchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-Prior to the start of regular production the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2 and 4.2.3. The first article sample shall consist of 15 complete sets of parts and sub-assemblies (i.e. 15 each of every component and sub-assembly, including at least three (3) items from each cavity used to produce plastic parts and be so identified as to permit tracing each item to the cavity in which it was molded) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including a certificate of conformance for materials. A sample containing known defects will not be submitted

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unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all of the examinations and tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly fails to comply with the applicable requirements, the first article sample shall be rejected. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

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4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraphs of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Body (see dwg. 7548051 covering a detail of 9207815).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Pitch diameter of plug thread maximum (max.)	Gage	01001
102.	Pitch diameter of external thread	Gage	01002
103.	Major diameter of external thread	Gage	01003
104.	Length of plug thread, minimum (min.)	Gage	01004
105.	Depth to bottom of cushion cavity, max	Gage	01005
106.	Depth to flash hole	Gage	01006
107.	Diameter of flash hole, min	Gage	01007
108.	Diameter of detonator cavity	Gage	01008
109.	Depth of detonator cavity	Gage	01009
110.	Diameter of body at rear end	Gage	01010
111.	Cuts, splits, or distortion	Visual	01011
Minor:	AQL 4.0 percent		
201.	Pitch diameter of plug thread min	Gage	01012

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202.	Diameter of plug activator counterbore	Gage	01013
203.	Depth of plug activator counterbore	Gage	01014
204.	Diameter of cavity below cushion	Gage	01015
205.	Depth to bottom of cushion cavity, min	Gage	01016
206.	Depth from external thread to rear end, max	Gage	01017
207.	Total length	Gage	01018
208.	Length from shoulder to rear end	Gage	01019
209.	Length from shoulder to external thread, max	Gage	01020
210.	Chamfer or radius missing	Visual	01021
211.	Evidence of poor workmanship, (see 3.3)	Visual	01022

4.3.2.2 Cup, (see dwg. 7548052 covering a detail of dwg. 9207815).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Inside diameter	Gage	02001
102.	Depth to bottom of cup cavity	Gage	02002
103.	Crack, split or distortion	Visual	02003
Minor:	AQL 1.50 percent		
201.	Outside diameter	Gage	02004
202.	Total length	Gage	02005
203.	Radius missing	Visual	02006
204.	Evidence of poor workmanship, (see 3.3)	Visual	02007

4.3.2.3 Plug, activator (see dwg. 7548050 covering a detail of dwg. 9207815).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 4.0 percent		

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201.	Pitch diameter of external thread	Gage	03001
202.	Major diameter of external thread	Gage	03002
203.	Length of thread, min	Gage	03003
204.	Outside diameter	Gage	03004
205.	Length to flange	Gage	03005
206.	Thickness of flange	Gage	03006
207.	Total length	Gage	03007
208.	Radius or chamfer missing	Visual	03008
209.	Crack, split, or distortion	Visual	03009
210.	Slot missing	Visual	03010
211.	Thread damaged	Visual	03011
212.	Evidence of poor workmanship, (see 3.3)	Visual	03012

4.3.3 Testing.-There are no applicable testing provisions.

4.3.4 Inspection Equipment.-Inspection equipment required to perform the inspections prescribed in this specification is identified in the Examination (4.3.2) and Test Method (4.4) paragraphs herein. See 6.3 for details concerning responsibilities for inspection design and approval.

4.3.5 Dimensional control of molded parts.-In place of normal sampling and inspections associated with the Classification of Defects, and after a curing time for the parts has been approved (see Note 1) a sample of three (3) parts (as molded) from each cavity shall be fully inspected dimensionally to quality a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraphs 4.3.2.1 through 4.3.2.3 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the three (3) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. All cavities removed from production because of some fault, may after reworking, be returned to production providing they pass the qualification test above. The contractor may request

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a change of inspection frequency providing he presents objective evidence to the contracting officer to substantiate the request. Contractor design of gages and test equipment required to perform the inspections listed herein shall be forwarded to Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-A-M for approval prior to manufacture of equipment. The noted subparagraphs identify those items and minimum inspections subject to the requirements of this paragraph.

NOTE 1. - In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to Picatinny Arsenal, Dover, New Jersey 07801 ATTN: SMUPA-QA-A-M. If any new change in material, cycle time, a reworked cavity, or a new cavity is used, a new curing time shall be established and approved.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.-Unless otherwise specified by the procuring activity, all packing and marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other

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inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or pro-

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curing the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SMUPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1345-A-024

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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